

Subject Description Form

Subject Code	DSAI1104
Subject Title	AI and Data Analytics Basics
Credit Value	1
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject introduces students to the basic concepts and techniques of artificial intelligence and data analytics (AIDA), and illustrates the application of AIDA in various industries in real life. The objective is to equip students with essential knowledge in statistics and computational tools, and to prepare them with fundamental AIDA skills, which are pivotal to maintaining and advancing Hong Kong's status as an international financial, innovation, and technology centre.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Demonstrate an understanding of the fundamental concepts of AIDA. - Demonstrate an understanding of mathematics and statistics knowledge behind AIDA. - Acquire basic skills in using AIDA technologies and applications. - Appreciate the importance of AIDA in the local and global community. - Analyse case studies to illustrate how AIDA are applied, with a focus on finance, innovation, and technology relevant to Hong Kong.
Subject Synopsis/ Indicative Syllabus	<u>General introduction to Data Science</u> Big data, data types, data collection, data analytics, selected case studies relevant to Hong Kong <u>Foundation Statistics</u> Descriptive statistics, measures of central tendency and dispersion, measures of association, linear regression <u>General introduction to Artificial Intelligence</u> Concept of machine learning, neural network, algorithms for classification via Python API such as scikit-learn

	<u>Application</u> Applications of artificial intelligence and data analytics in specific topics such as finance analytics, regression analysis, prediction, healthcare, smart city development, etc.						
Teaching/Learning Methodology	The subject will be delivered through lectures and labs sessions. The lectures will be conducted to introduce the basic AI and data analytics concepts, which are then reinforced by learning activities and demonstrations. Labs will be conducted to introduce practical techniques in computer programming, and to do hands-on machine learning exercise.						
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	1. Quizzes	10 %	√	√		√	
	2. Hands-on programming	10 %	√		√	√	
	3. Participation	20%	√	√	√	√	√
	4. AIDA case study report	60 %	√	√	√	√	√
	Total	100 %					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: (1) Quizzes / Programming: This will allow the instructor to observe and assess individual student’s achievement of a particular learning outcome based on the coverage of the assigned problems. Some of the assignment or questions might require students to self-learn with reading references. (2) Participation: The class attendance and interaction will allow the instructor to observe and assess students’ overall understanding of AIDA knowledge. (3) Case study report: Students are required to submit an individual report. This should include a case study regarding the application of AIDA in their chosen discipline (finance, social science, marketing, healthcare, etc). This will help students to develop problem analysis and design skills; and allow the instructor to assess students’ overall understanding of AIDA knowledge.						

Student Study Effort Expected	Class contact:	
	▪ Lectures/ Labs/ Seminars	13 Hrs.
	Other student study effort:	
	▪ Self-study / Reference reading	26 Hrs.
	Total student study effort	39 Hrs.
Reading List and References	Raschka, S. and Mirjalili, V. (2019). Python Machine Learning: Machine Learning and Deep Learning with Python, Scikit-Learn, and TensorFlow (3rd Edition). Packt Publishing. Adhikari, A. and DeNero, J. (2019). Computational and Inferential Thinking: The Foundations of Data Science. GitBook. Davy, C., Meysman, A. D. B., and Ali, M. (2016). Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools. Manning Publications. Utts, J. M. and Heckard, R. F. (2015). Mind on Statistics (5th edition). Cengage Learning.	